SEMINAR REPORT

A seminar Report on

AI-Based Crop Disease Detection System

Submitted in partial fulfillment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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FOR THE ACADEMIC YEAR : 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING S.S.V.P.S's

Bapusaheb Shivajirao Deore Polytechnic, Dhule (Institute Code 0059)

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Polytechnic, Dhule (Inst.Code: 059) has completed the term work satisfactorily in Course **Seminar Presentation and Project Initiation (Course Code: 315003)** for the academic year **2026 to 2027** as prescribed in the curriculum.

Place : Dhule

Date : .....

Signature:

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Subject Teacher

Signature:

Name :

Head of the Department

Seal of Institution

TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Review	2
3	Information/Chapters	3
4	Advantages and Disadvantages	8
5	Conclusion	9
6	Project Initiation :	10
	6.1) Description of problem statement.	10
	6.2) Scope and objectives.	10
	6.3) Stake holder	11
	6.4) Platform/ Equipment/ Resources identification.	12
7	Seminar Presentation	
	Bibliography	
	References	

ABSTRACT

The occurrence of crop diseases and the damage caused to agricultural crops create significant challenges in farming. Traditional methods of disease identification often depend on manual observation and farmers' experience, which can lead to delayed or incorrect identification of diseases. This may affect crop quality, reduce productivity, and create difficulties in taking timely action.

To address these challenges, this project presents an AI-Based Crop Disease Detection System, designed to identify crop diseases using leaf images. The system uses image processing and Artificial Intelligence techniques to analyse crop leaf photographs and identify visible disease symptoms. Based on the collected image, the system helps determine whether a leaf is healthy or affected by a disease. It can also display detection results to support better understanding of crop health.The proposed system aims to support early disease identification, reduce dependence on manual observation, and help farmers make better decisions. AI-Based Crop Disease Detection offers a practical approach to crop monitoring and can support efficient agricultural practices.

LIST OF FIGURES

Fig. No.	Title	Page No.
1.1	Working of AI-Based Crop Disease Detection	8
1.2	Architecture of AI-Based Crop Disease Detection	10

LIST OF TABLES

Table No.	Title	Page No.
2.1	Literature Review	5
6.1	Resources	15

Chapter1

INTRODUCTION

Agriculture plays an important role in the Indian economy and provides food and employment to a large part of the population. However, crop diseases are one of the major problems faced by farmers. These diseases can affect the growth of plants, reduce crop quality, and cause losses in agricultural production. Traditional methods of disease identification mainly depend on the visual observation and experience of farmers, which may sometimes lead to incorrect identification or delayed treatment.

The AI-Based Crop Disease Detection System is a technology-based solution designed to identify possible crop diseases by analysing images of plant leaves. It uses Artificial Intelligence, image processing, and deep learning techniques to examine leaf images and recognise visible symptoms of diseases. The system can help users identify whether a crop leaf appears healthy or affected by a disease included in the trained model.

The system works by accepting an image of a crop leaf from the user. The image is processed to improve its suitability for analysis and is then passed to a trained AI model. The model analyses the image patterns and generates a predicted result, which is displayed to the user. This process can make preliminary disease identification easier and help farmers understand the possible condition of their crops.

AI-Based Crop Disease Detection can be developed using technologies such as Python, OpenCV, TensorFlow, and Convolutional Neural Networks (CNN). These technologies support image processing and classification. The system can also include a simple user interface through which users can upload images and view the detection results.

The main objective of this system is to support early crop disease identification, reduce dependence on manual observation, and encourage the use of modern technology in agriculture. However, the accuracy of the result depends on image quality, the training dataset, and the capabilities of the model. Therefore, the system should be used as an assistance tool, and agricultural experts should be consulted when confirmation or treatment advice is required.

Chapter2

LITERATURE REVIEW

AI-Based Crop Disease Detection

1. Using Deep Learning for Image-Based Plant Disease Detection

Mohanty, Hughes, and Salathé (2016) studied the use of deep learning for identifying plant diseases from leaf images. Their research used the PlantVillage dataset and convolutional neural networks to classify healthy and diseased plants. The model learned visual patterns from leaf images and predicted the possible disease category. The study demonstrated the potential of Artificial Intelligence for automated plant disease identification. However, images captured under different lighting conditions and backgrounds can affect the performance of the model.

2. Deep Learning Models for Plant Disease Detection and Diagnosis

Ferentinos (2018) investigated deep learning models for detecting and diagnosing plant diseases from images. The models analysed visible symptoms on plants and classified images into different disease categories. The study showed that deep learning can support automatic plant disease identification. The results depend on the quality and variety of images, the training process, and the model selected for classification.

3. Comparative Study of Deep Learning Models for Plant Disease Identification

Too and other researchers (2019) compared different deep learning architectures for plant disease identification. Models such as VGG, ResNet, Inception, and DenseNet were evaluated for image classification. The study highlighted the importance of selecting a suitable architecture for accurate classification. However, model complexity and differences between training images and real-field images can create challenges

AI-Based Crop Disease Detection

1. Image Processing for Crop Disease Detection

Image processing is an important technique used to prepare crop leaf images for analysis. The image may be resized, converted into a suitable format, and normalised before it is given to the AI model. These operations help maintain consistency between input images and training images. Image processing supports the identification of visible patterns such as spots, colour changes, and damaged areas on leaves. However, poor image quality and unsuitable lighting may affect the detection result

2. Intelligent Load Scheduling for Energy Efficiency

Intelligent load scheduling focuses on operating electrical appliances at suitable times according to energy demand, electricity tariffs, and user requirements. Essential appliances can continue operating when needed, while flexible loads can be shifted to off-peak periods. This approach can reduce peak electricity demand, improve load distribution, and help lower electricity costs.

3. AI-Based Crop Monitoring and Disease Identification

AI-based crop monitoring uses image analysis and machine learning to support the identification of plant diseases. Users can capture or upload leaf images, and the system analyses them to generate a predicted result. Such systems can make preliminary disease screening more convenient for farmers and agricultural researchers. However, the prediction should be treated as an initial indication because some diseases may have similar visible symptoms.

AI-Based Crop Disease Detection

1. Early Identification of Crop Diseases

Early identification helps farmers recognise possible disease symptoms before the problem spreads further. AI-based systems analyse leaf images and compare their visual patterns with those learned during training. This can help users identify possible diseases and decide whether further inspection is required. The usefulness of the result depends on the diseases supported by the model and the quality of the input image

2. Automated Disease Classification Using Leaf Images

Load categorization classifies electrical appliances according to their importance and flexibility. Essential loads operate according to user requirements, whereas flexible loads can be scheduled for suitable periods. This technique helps manage peak demand and improve energy efficiency. It is particularly useful for residential energy management under different electricity tariff structures.

3. Crop Disease Detection Through Mobile and Web Applications

Mobile and web applications can provide an easy interface for AI-based crop disease detection. Users can upload photographs of crop leaves and view the predicted result on their devices. A web-based system can connect the user interface with an image-processing module and a trained classification model. Such applications can improve accessibility and simplify the use of AI technology in agriculture. However, internet access may be required for systems that depend on an online server.

AI-Based Crop Disease Detection and Classification

1. Deep Learning-Based Plant Disease Recognition

Deep learning-based plant disease recognition uses trained neural networks to analyse crop leaf images. The model learns visual features from a collection of labelled images and uses them to predict the disease category of a new image. This approach supports automated image classification and can assist in preliminary crop disease identification.

2. Image-Based Disease Classification Using CNN

CNN-based classification analyses patterns such as leaf spots, discolouration, and visible damage. The image is preprocessed and passed to the trained model, which generates a predicted class. CNNs are suitable for image-related tasks, but their performance depends on the diversity of training images and the conditions under which new images are captured.

3. AI-Assisted Crop Health Monitoring

AI-assisted crop health monitoring uses image analysis to provide information about the possible condition of plants. The system can display a prediction and help users identify leaves that may require further examination. This approach supports modern agricultural practices, but expert confirmation may be necessary before selecting a disease treatment.

Research Papers

1. Mohanty, S. P., Hughes, D. P., and Salathé, M. (2016). “Using Deep Learning for ImageBased Plant Disease Detection.” Frontiers in Plant Science, 7, 1419.
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Table No 2.1: Literature review

Sr. No	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	Image-Based Plant Disease Detection	Deep Learning, CNN	PlantVillage leaf images	To identify plant diseases using leaf images	Performance may vary with realfield images
2	Deep Learning-Based Disease Diagnosis	CNN, Image Classification	Healthy and diseased plant images	To classify different plant diseases	Depends on image quality and training data
3	Comparative Study of Disease Identification	VGG, ResNet, Inception, DenseNet	PlantVillage dataset	To compare deep learning models for disease classification	Some models require greater computing resources

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	Mobile-Based Plant Disease Detection	Android, Python, CNN	Plant Leaf Images	To detect plant diseases using a mobile application.	Performance depends on image quality and device capability.
2	Transfer Learning-Based Disease Detection	VGG16, ResNet50 MobileNet	Plant Disease Image Dataset	To improve disease classification using pretrained models.	Requires sufficient computational resources for training.
3	Real-Time Plant Disease Recognition	Deep Learning, CNN, OpenCV	Real-Time Plant Leaf Images	To identify plant diseases from leaf images in real time.	Lighting conditions and background noise may affect accuracy.

Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	Image-Based Plant Disease Detection	Deep Learning, CNN	PlantVillage Leaf Images	To identify plant diseases using leaf images..	Performance may vary with realfield images.
2	Deep Learning-Based Disease Diagnosis	CNN, Image Classification	Healthy and Diseased Plant Images	To classify different plant diseases.	Depends on image quality and training data.
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Sr. No.	Approach Type	Technology Used	Dataset	Key Objective	Limitations
1	AI-Based Plant Disease Classification	Artificial Intelligene, CNN	Plant Diseae Image Dataset	To detect plant diseases automatic	Accuracy depends on the quality of training data.
2	Machine Learning-Based Disease Prediction	Machine Learning, Feature Extraction	Plant Leaf Image Dataset	To classify plant diseases using extracted features.	Feature selection can affect performane.
3	Deep Learning-Based Plant Disease Recognition	CNN, Python, TensorFlow	Healthy and Diseased Leaf Images	To improve disease recognition accuracy.	Training may require high computatial resources.

INFORMATION/CHAPTERS

1. **Overview of AI-Based Crop Disease Detection System:**

AI-Based Crop Disease Detection System is a technology designed to identify plant diseases using Artificial Intelligence (AI) and image processing techniques. It analyzes images of crop leaves to detect disease symptoms such as spots, discoloration, and unusual patterns. The main goal of this system is to help farmers identify crop diseases at an early stage, reduce crop losses, and improve agricultural productivity.

2. **Crop Disease Monitoring:**

Crop disease monitoring involves observing and analyzing plants to identify signs of diseases. In an AI-Based Crop Disease Detection System, farmers can capture images of crop leaves using a smartphone or camera. The system processes these images and uses a trained AI model to identify whether the plant is healthy or infected. This information helps farmers understand the condition of their crops and take appropriate measures to control the spread of diseases.

3. **Need for Intelligent Crop Disease Detection:**

Traditional crop disease detection methods often depend on farmers' experience and manual observation, which may lead to incorrect identification or delayed treatment. An AI-Based Crop Disease Detection System uses automated image analysis to detect diseases quickly and efficiently. It helps farmers identify potential plant diseases at an early stage, reduce unnecessary pesticide usage, minimize crop damage, and improve crop quality and yield.

1. Working Principle:

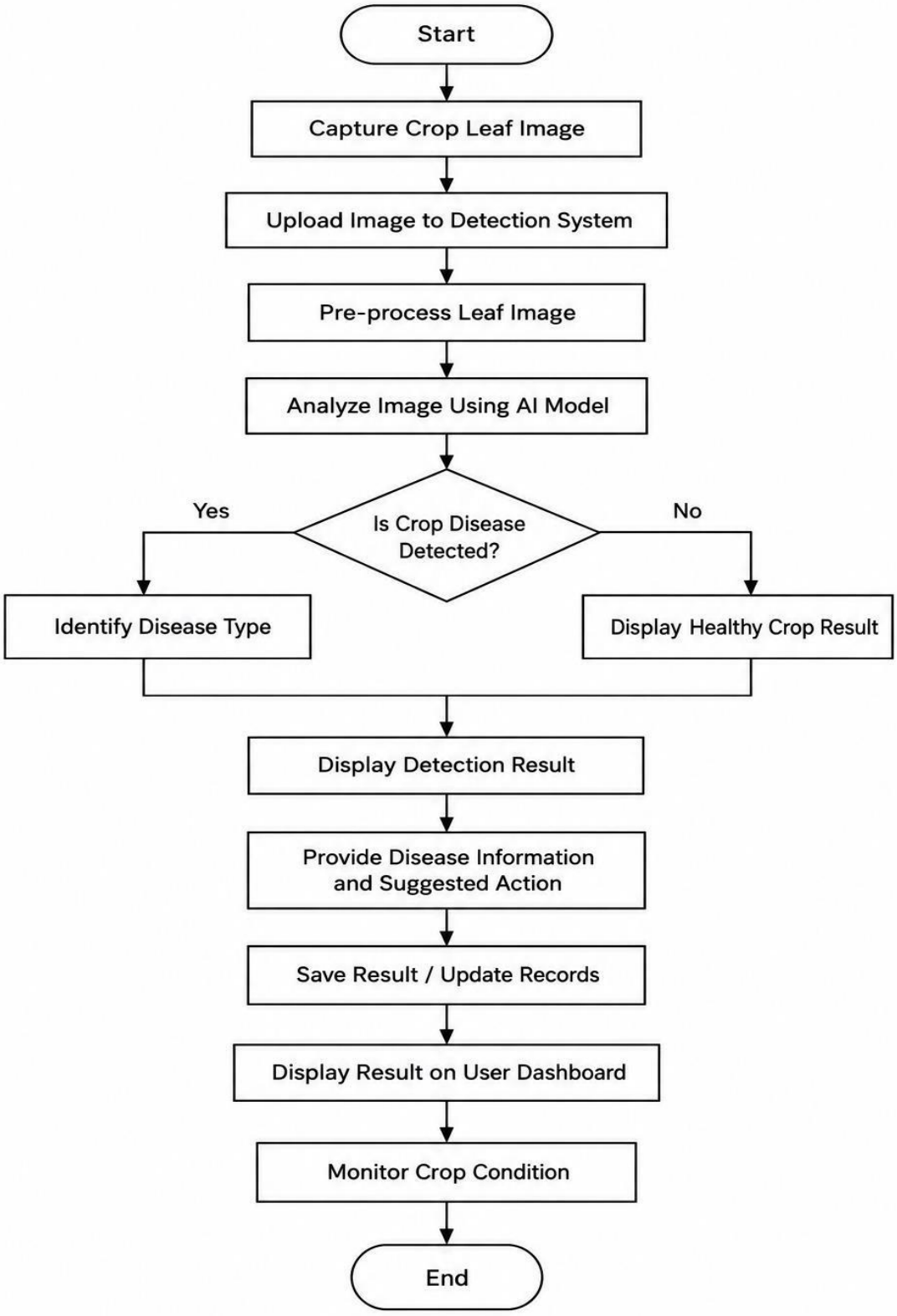


Fig 3.1: Working of AI-Based Crop Disease Detection System

The working of the AI-Based Crop Disease Detection System consists of the following steps:

1. Start: The process begins when the user opens the AI-Based Crop Disease Detection System to identify crop diseases.
2. Capture Crop Leaf Image: The user captures an image of the crop leaf using a smartphone or camera to examine visible signs of disease.
3. Upload Image to Detection System: The captured image is uploaded to the system for further processing and disease detection.
4. Pre-process Leaf Image: The system processes the uploaded image by adjusting its size, quality, and format to make it suitable for AI analysis.
5. Analyze Image Using AI Model: The AI model analyzes the leaf image and identifies patterns, spots, discoloration, and other features associated with crop diseases.
6. Decide Whether Crop Disease Is Detected: The system checks the AI model's prediction. If a disease is detected, the system identifies the possible disease type. If no disease is detected, it displays the healthy crop result.
7. Identify Disease Type: If the crop is infected, the system displays the predicted disease name based on the patterns identified in the leaf image.
8. Display Detection Result: The system displays the result, indicating whether the crop is healthy or affected by a particular disease.
9. Provide Disease Information and Suggested Action: The system provides relevant information about the detected disease and suggests suitable preventive or control measures.
10. Save Result and Update Records: The detection result can be stored in the system database for future reference and crop health monitoring.
11. Display Result on User Dashboard: The system displays the detection results and related information on the user dashboard for easy access.
12. Monitor Crop Condition: Farmers can regularly upload new crop leaf images to monitor plant health and identify possible diseases at an early stage.
13. End: The process completes after the disease detection result is displayed and the required information is provided to the user.

2. System Architecture and Main Components:

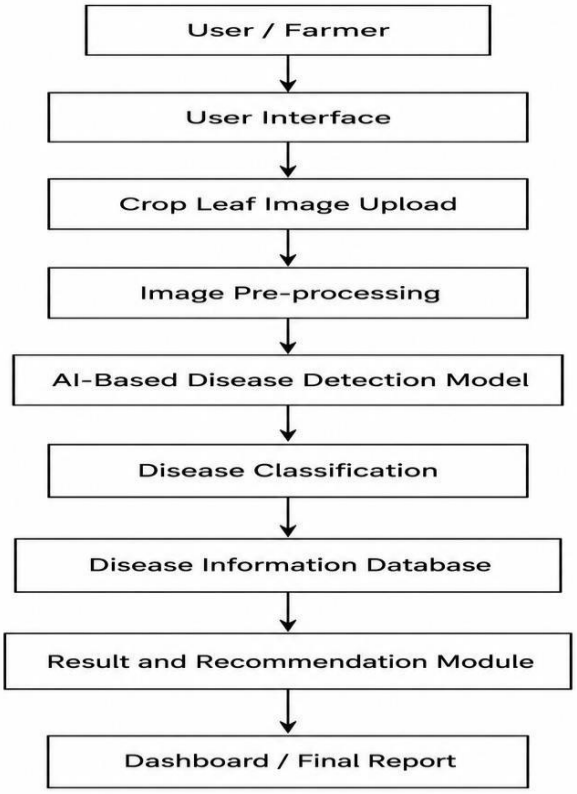


Fig 3.2: Architecture of AI-Based Crop Disease Detection System

The main components of the AI-Based Crop Disease Detection System include the User Interface, Crop Image Acquisition Module, Image Pre-processing Module, AI-Based Disease Detection Model, Disease Classification Module, Disease Information Database, and Result and Recommendation Module. These components work together to analyze crop leaf images, identify possible plant diseases, classify the detected diseases, and display detection results with suitable preventive measures. The system helps farmers monitor crop health, detect diseases at an early stage, and reduce crop losses.

1. Technologies and Techniques Used:

The AI-Based Crop Disease Detection System can be developed using technologies such as Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, image processing, and database management systems. Image acquisition techniques are used to capture crop leaf images, while image pre-processing prepares the images for analysis. A trained AI model, such as a Convolutional Neural Network (CNN), analyzes the images and identifies possible crop diseases. Database systems store disease information and detection records, while a user interface displays the results and suggested preventive measures. The final technology selection depends on the project implementation, available resources, and dataset.

2. Example of Crop Disease Detection:

Consider a farm where tomato, potato, and other crops are grown. The AI-Based Crop Disease Detection System analyzes images of crop leaves to identify signs of diseases such as spots, discoloration, or unusual patterns. If a disease is detected, the system displays the possible disease name and provides relevant information to the farmer. This helps farmers identify crop diseases early, take suitable preventive measures, and reduce crop losses.

3. Crop Disease Monitoring and Detection:

The system accepts crop leaf images through a smartphone camera or other imaging devices and analyzes them using image processing and an AI model. The detection results are displayed on a user dashboard, allowing farmers to understand the health condition of their crops. If a possible disease is detected, the system can provide disease information and suggested control measures. This process helps improve crop monitoring, supports timely decision-making, and promotes better agricultural productivity.

ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Early Disease Detection: Helps identify possible crop diseases at an early stage by analyzing crop leaf images using Artificial Intelligence.

2. Improved Crop Monitoring: Supports regular monitoring of plant health and helps farmers understand the condition of their crops.

3. Reduced Crop Losses: Early identification of diseases helps farmers take suitable preventive measures and minimize crop damage.

4. Quick and Automated Detection: Analyzes crop images automatically and provides detection results in less time than manual inspection.

5. Easy Access to Information: Displays the possible disease name, related information, and suggested preventive measures through a user interface.

4.2 Disadvantages:

1. Dataset Dependency: The accuracy of the system depends on the quality, quantity, and variety of crop images used to train the AI model.

2. Image Quality Issues: Blurry images, poor lighting, and unclear leaf symptoms may affect disease detection accuracy.

3. Technical Complexity: Developing and training an AI model requires suitable technical knowledge and programming skills.

4. Limited Disease Recognition: The system may not identify diseases that are absent from its training dataset or recognize every disease under real-world conditions.

5. Internet and Device Requirements: Depending on the implementation, the system may require a smartphone, camera, internet connection, or sufficient computing resources

CONCLUSION

The AI-Based Crop Disease Detection System provides an efficient approach to identifying and monitoring crop diseases using Artificial Intelligence and image processing techniques. It analyzes crop leaf images, identifies possible disease symptoms, and displays the detection results to help farmers understand the health condition of their crops.

By combining image acquisition, image pre-processing, AI-based disease classification, database management, and result display, the system aims to support early disease detection and reduce crop losses. Although challenges such as dataset quality, image clarity, development costs, and technical complexity remain, proper design and implementation can improve its effectiveness.

In conclusion, the AI-Based Crop Disease Detection System can serve as a useful solution for detecting possible crop diseases, supporting timely agricultural decisions, improving crop monitoring, and promoting better agricultural productivity.

PROJECT INITIATION

a) Description of Problem Statement:

Crop diseases are a major challenge in agriculture because they can reduce crop quality, decrease agricultural production, and cause financial losses to farmers. Identifying crop diseases at an early stage is important for preventing their spread and protecting healthy plants. However, conventional disease detection methods often depend on manual observation and farmers' experience, which may lead to incorrect identification or delayed treatment.

Traditional methods of crop disease identification can be time-consuming and may require expert knowledge to recognize different disease symptoms. Therefore, there is a need for a system that can analyze crop leaf images, identify possible diseases, and provide useful information to farmers for taking appropriate preventive measures.

The proposed AI-Based Crop Disease Detection System aims to address this problem by using Artificial Intelligence, image processing, and deep learning techniques. It analyzes uploaded crop leaf images, identifies possible disease symptoms, and displays the detection results to the user. The objective is to support early disease detection, reduce crop losses, and improve crop health monitoring**.**

b) Scope of the Project:

The scope of the AI-Based Crop Disease Detection System is to develop an intelligent system that can identify possible crop diseases by analyzing images of plant leaves. The system can use image processing techniques and a trained AI model to classify crop images as healthy or affected by a particular disease. It can also display disease-related information and suggested preventive measures to help farmers understand the detected problem.

The project focuses on simplifying crop disease identification, supporting early detection, reducing dependence on manual inspection, and improving agricultural decision-making. The system can be designed as a web application or mobile application, depending on the available resources and project requirements. Its effectiveness depends on the quality of the input images, the training dataset, and the accuracy of the AI model.

Objectives of the Project:

1. To identify possible crop diseases by analyzing crop leaf images using Artificial Intelligence.
2. To collect and process crop images for disease detection and classification.
3. To distinguish between healthy and diseased crops based on the trained AI model.
4. To provide quick and automated crop disease detection results.
5. To display disease-related information and suggested preventive measures to help farmers.
6. To reduce crop losses, support early disease detection, and improve agricultural productivity.

c) Stakeholders:

The stakeholders of the AI-Based Crop Disease Detection System are the individuals and groups who use, develop, maintain, or benefit from the system.

1. End Users: Farmers who use the system to upload crop leaf images, identify possible diseases, and obtain relevant crop health information.
2. Developers: The team responsible for designing, developing, testing, and maintaining the AI-Based Crop Disease Detection System.
3. Agricultural Experts: Professionals who provide knowledge about crop diseases, symptoms, prevention, and suitable control measures.
4. System Administrators: Individuals responsible for managing the system, maintaining the database, and ensuring its proper operation.
5. Agricultural Organizations: Agricultural institutions, research centers, and farming organizations that may use the system to support crop health monitoring and disease management.

d) Platform / Equipment / Resources:

1. Platform:

The AI-Based Crop Disease Detection System can be developed on the Windows or Linux operating system using suitable programming languages and development tools. Python can be used for AI model development and image processing, while libraries such as TensorFlow, Keras, or OpenCV can be used for crop disease detection and image analysis. A code editor such as Visual Studio Code can be used for development and testing. The system may use a trained AI model and a database to identify crop diseases, store detection results, and display disease-related information to users.

2. Equipment:

· **Computer or Laptop**: For developing, training, and testing the application.

· **Processor:** A standard modern processor suitable for Python programming and AI model development.

· **RAM:** Minimum 8 GB recommended; additional memory may be required for training larger AI models.

· **Internet Connection:** Required for downloading libraries, accessing datasets, and obtaining development resources.

· **Storage:** Sufficient space for source code, crop image datasets, required libraries, and trained AI model files.

· **Camera or Smartphone:** For capturing crop leaf images and providing input to the disease detection system.

3. Resources:

Resources	Name	Purpose
Operating System	Windows 11	Host and run development tools, editors, and local testing environment.
Programming Language	Python 3.11	Primary language used to implement application code, scripts, and tests.
Code Editor	Visual Studio Code	Code editing, debugging, and extension support for development.
Iot Development Tool	Arduino IDE	Used to write, upload, and test programs on compatible microcontrollers.
Sensor Data Processing	Python Libraries	Used to process voltage, current, and power consumption data collected from sensors or monitoring devices.
Database	MySql	Used to store, organize, and retrieve electricity consumption records and related system data.
Data Visualization	Matplotlib	Used to display energy consumption patterns and generate graphical reports.

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrolment Number	Seat Number	Name of student
1	24610960185	344504	Chaitali Sanjay Chaudhari
2	24610960214	388522	Yogita Jitendra Khairnar
3	24610960239	388547	Kalyani Santosh Patil
4	24610960240	388548	Kashish Atul Patil

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388504	388522	388547	388548
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign : Name : ----------

(Course Expert/s)